

Learning-Based Scene Selection for Autonomous Multi-Camera Video Editing

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Abstract—In recent years, the ability to easily shoot multi-camera videos has become possible, and creators are burdened by the large amount of time spent editing these data. This study proposes a learning-based scene-selection framework for autonomous multicamera video editing. This method learns the editor's "selection" from an extremely limited training dataset (33 scenes). Our framework consists of three stages: (1) extracting semantic features from video frames using OpenAI CLIP, (2) learning the editor's binary "Adopt/Reject" preferences using a Support Vector Machine (SVM), and (3) implementing a novel editing logic that applies this scene-level score to a frame-level selection process. As a result of the experiment, when the video work autonomously generated by this framework was evaluated by a human, 79.66% of the scenes were judged as "Usable" (Usable Scene Ratio). Furthermore, this AI selection was statistically significant ($p < 0.0001$) compared to random selection. This research demonstrates that, even with extremely limited data, it is possible to autonomously generate practical-quality video edits that reflect human sensibilities.

Keywords—Automatic video editing, Scene detection, Available scene detection, Video generation

I. INTRODUCTION

The rapid advancement and widespread adoption of generative AI and high-quality smartphones have made multicamera video shooting and automated content generation accessible to virtually anyone. Platforms such as YouTube host an ever-growing, searchable, and viewable body of video. Even in this era of generative AI, creators are expected to deliver an immense volume and variety of video content in real-time.

However, this explosion of content has created a significant bottleneck in the production pipeline and editing process. The burden of editing multicamera footage remains a substantial barrier for creators. This bottleneck is particularly evident in the traditional multicamera editing workflow, which involves two major phases with significant cognitive load.

The multicamera editing process can be divided into two phases. In the first phase, scene segmentation, the editor watches each camera's raw footage, often for hours, to identify semantic boundaries and divide continuous footage into discrete scenes or shots. The second phase, scene selection, is significantly more cognitively demanding. Here, the editor must horizontally compare all scenes across synchronized cameras and decide, for each moment, which camera's scene should be adopted and which should be rejected to construct the final narrative.

For example, a 30-minute recording captured by six cameras requires the creator to review at least 180 min of raw material, not just for segmentation, but also to mentally juggle the combinatorial possibilities of the "Scene Selection" phase. This manual selection process is not only time-consuming but also relies heavily on the editor's implicit "sense" or "style," which is difficult to articulate or delegate.

This challenge is amplified in the "Post-Generative Editing" era. As AI systems generate vast quantities of video material, the need for an automated system that can filter, select, and refine this material based on human-like judgment has become paramount.

To bridge this gap, we propose a learning-based Scene Selection method for autonomous multicamera video editing, focusing specifically on automating the burdensome scene selection phase. This method learns the creator's "selection sense" from existing studies. We believe that this is the core of our study.

Our framework was uniquely designed to operate under real-world constraints. It employs semantic feature extraction via OpenAI CLIP to comprehend video content beyond simple pixel data. In addition, it utilizes preference learning through a Support Vector Machine (SVM), a robust classifier specifically selected for its ability to learn effectively from extremely limited training data—only 33 scenes—available from a single editor. Finally, it introduces a novel scene-to-frame selection logic that translates abstract scene-level preference scores into concrete frame-by-frame editing decisions, thereby elegantly addressing the challenge of asynchronous scene boundaries.

By automating this labor-intensive stage, creators can offload the editing workload and shorten production cycles. This system transforms large raw datasets into polished works by learning and applying the judgment patterns of the editor.

The main contributions of this study are threefold:

1. We introduce an integrated framework for autonomous multi-camera video editing that combines feature extraction (CLIP), a preference learning engine (SVM), and a novel scene-to-frame selection logic. We emphasize that this framework is specifically designed for limited-data-driven scenarios, which is a common real-world challenge.
2. We built an experimental testbed and demonstrated that even from an extremely limited training dataset (33

scenes), our framework can autonomously generate a complete video. When this AI-generated product was evaluated by a human editor, 79.66% of the scenes were deemed 'usable' (Usable Scene Ratio). This result significantly outperformed that of random selection ($p < 0.0001$).

3. We discuss the implications of our approach for "Post-Generative Editing" workflows and identify the integration of audio information, a critical component revealed by our qualitative analysis, as the next step for future work.

The remainder of this paper is organized as follows. Section II reviews the relevant literature. Section III describes the proposed method in detail. Section IV presents the experimental settings and results. Section V offers a discussion and Section VI concludes the paper.

II. RELATED WORKS

The fields of automatic video editing and multicamera processing have evolved significantly over the past decade, driven by advances in machine learning and the growing demand for efficient content creation. This section reviews the relevant literature across five key areas that inform our proposed learning-based scene selection method for autonomous multicamera video editing.

The transition from manual to automatic video editing has been facilitated by data-driven approaches that learn editing patterns from professionally edited content. Podlesnyy et al. [1] pioneered the use of learned editing controllers, employing ImageNet-based visual features and reinforcement learning to train sequence controllers for cutting and pacing decisions. Their work demonstrated that imitation learning from professional editors could produce more cinematic results than traditional rule-based approaches.

The development of large-scale editing datasets has accelerated the progress in this domain. The Anatomy of Video Editing dataset [2] provides approximately 196,000 sampled shots with 1.5 million cinematography annotations, enabling systematic evaluation of AI-assisted editing tasks. This benchmark has proven instrumental in moving beyond visual effects to learn actual editing decisions and shot attributes of videos.

Recent advances in video summarization have emphasized conditional modeling approaches that incorporate non-visual factors such as interestingness, representativeness, and storyline consistency [3]. These methods use conditional attention modules to fuse multimodal cues and produce summaries that are better aligned with human preferences. Huynh-Lam et al. [4] further enhanced the summarization quality using context-aware pipelines that captured the semantic relationships between video segments.

However, existing automatic editing approaches predominantly focus on single-stream video processing or shot-level selection without explicitly addressing the challenges of multi-camera environments. The proposed method extends these foundations by learning scene-level selection policies across synchronized camera views, thereby addressing a significant gap in the current automatic editing research.

Robust scene boundary detection is a critical foundation for scene-level video analysis and editing. Recent advances have moved beyond traditional histogram-based methods to deep learning approaches that capture semantic transitions in scenes.

Tan et al. [5] introduced Video Scene Montage for Boundary Detection (VSMBD), a self-supervised approach that synthesizes pseudo scene boundaries to train detection models without manual annotations. Their method decouples foreground and background analyses, enabling a more robust detection of semantic scene transitions than shot-level change detection.

Deep learning architectures have also been adapted for shot boundary detection, with Brotons et al. [6] proposing 3D depth-wise convolutions combined with visual attention. Their approach achieved improved accuracy in detecting abrupt transitions while maintaining the computational efficiency. Similarly, RajaSuguna et al. [7] developed adaptive threshold methods using color histogram differences and SURF feature descriptors for candidate segment identification.

Optical flow-based approaches have shown promise for temporal segmentation, with Hato [8] demonstrating effective shot transition detection using motion analysis. These methods complement appearance-based approaches by capturing temporal dynamics that may not be evident in static analysis.

Although these advances provide robust scene detection capabilities, most existing methods focus on single-camera applications. The proposed approach leverages these segmentation techniques and extends them to handle synchronized multi-camera streams, thereby enabling consistent scene boundaries across multiple viewpoints.

The processing of multi-camera video data has attracted increasing attention owing to the proliferation of multi-view recording systems. Research in this area has focused on view selection, camera switching, and content-aware processing.

Hou et al. [9] developed learning-based camera-view selection methods that efficiently process multiview data with minimal computational requirements. Their approach uses attention mechanisms to identify the most informative viewpoints for downstream tasks, demonstrating significant improvements in both accuracy and efficiency compared with exhaustive processing methods.

Language-guided view selection has emerged as a promising direction, with Majumder et al. [10] proposing weakly supervised methods that use natural language descriptions to guide camera selection in multi-view videos. Their approach enables more intuitive control over the view selection policies while reducing the need for extensive manual annotations.

Context-based camera selection was explored by Lefevre et al. [11], who developed algorithms for automatic switching between multiple video streams based on the content analysis. Their method considers factors such as motion, face detection, and audio cues to determine optimal camera transitions, although it operates at the shot level rather than at the scene level.

Patent literature has also documented practical approaches to multiview video processing, with Ranjeet et al. [12]

describing systems for automatic view selection based on content quality metrics and user preference. These industrial applications highlight the practical importance of automated multicamera processing.

The proposed method builds upon these multi-view processing advances by introducing scene-level selection policies that consider both content quality and editing aesthetics, moving beyond purely technical metrics to incorporate learned editorial preferences.

Machine learning has become increasingly prevalent in video analysis tasks, with deep learning approaches showing particular promise in solving complex video understanding problems.

Sports video analysis has served as a testbed for advanced ML techniques, with Host et al. [13] applying deep neural networks to analyze the movements and activities of a handball player. Their work demonstrated the effectiveness of CNN-based approaches for extracting meaningful patterns from complex video data. Similarly, Liu et al. [14] developed deep learning methods for football game video analysis, demonstrating how domain-specific models can achieve superior performance compared with generic approaches.

Attention mechanisms are particularly effective for video analysis. Ruiye [15] employed BiLSTM fusion with attention mechanisms for volleyball training video classification, achieving improved accuracy in temporal pattern recognition. These attention-based approaches provide interpretable insights into the temporal segments that contribute most to the classification decisions.

Recent studies have explored the integration of vision-language models for video editing tasks. Hu et al. [16] proposed reinforcement learning-based automatic video editing using pre-trained vision-language models, demonstrating how large-scale pre-training can be leveraged for downstream editing applications.

The proposed method incorporates these ML advances by using supervised learning to capture editorial preferences from annotated multi-camera datasets while leveraging attention mechanisms to focus on the most relevant scene characteristics for selection decisions.

Content-aware video processing has evolved to consider perceptual quality, semantic content, and user preferences in automated selection systems. This research area is particularly relevant to the proposed scene selection approach.

Neural atlas graphs have been proposed for dynamic scene decomposition and editing [17], which enables sophisticated content analysis and manipulation. These methods provide fine-grained control over video content while maintaining temporal consistency across edits.

Storyboard-based editing systems have demonstrated the value of a high-level understanding of content for automatic video editing. Chen [18] developed storyboard-based summarization that considers narrative structure and content flow, achieving more coherent edited results compared to purely statistical approaches.

Quality assessment metrics have also evolved to better capture human perceptual preferences. Recent studies have incorporated learned quality models that consider factors such as composition, motion characteristics, and semantic content relevance.

However, most existing content-aware approaches focus on single-camera scenarios or post-production editing rather than real-time multi-camera selections. The proposed method addresses this gap by learning content quality assessment specifically for multicamera environments, enabling real-time scene-level selection based on learned editorial preferences.

The literature review highlighted several key gaps that the proposed method aims to address. Existing studies have largely focused on shot-level analysis or single-camera processing, with limited exploration of scene-level selection across synchronized multicamera streams. Moreover, most current approaches depend on technical quality metrics rather than learning editorial preferences from professional content creators. Another shortcoming is the lack of integrated workflows; many methods treat scene detection, view selection, and editing as separate tasks instead of combining them into a cohesive editing framework. Finally, scalability remains a major issue, as many existing techniques require extensive manual annotation or are constrained to specific domains, whereas real-world applications demand generalizable methods capable of adapting to diverse types of content.

The proposed learning-based scene selection method addresses these gaps by providing an integrated framework that learns editorial preferences from annotated multi-camera datasets and applies this knowledge to automate scene-level selection decisions across diverse content types.

III. METHOD

Our proposed method generates a single work from multi-camera data in two phases: a Training Phase and a Generation Phase. The core of our method is to learn an editor's "Adaptability" preference from a small set of examples, and then apply that learned knowledge to a novel frame-by-frame selection logic.

A. Overview of Our Proposed Method

An overview of the proposed method is shown in Fig. 1. The method consists of two phases: the Training Phase (multi-camera video data annotation phase) and the Generation Phase (automatic multi-camera video data editing and generation phase) to efficiently generate a single work. In the Training Phase, the existing edited footage was used. First, an automatic scene-division function (III. B), the video was split into scenes. These scenes are then labeled (adopted/rejected) by the human editor to construct a Usage Scene Label Database. This database was the input for the Available Scene Training Function (III.C), which trained our SVM preference model. In the Generation Phase, a new, unseen multicamera footage is input by the user. The footage was similarly split into scenes using the Automatic Scene Division Function (III.B). The trained Available Scene Detection Function (III.D) evaluates each new scene and outputs an "Adaptability Score". Finally, the Automatic Video Editing Function (III.E) uses these scores to select the best camera frame by frame, assembling the adopted scenes chronologically to

produce the final video. This workflow significantly reduces the manual burden of the scene selection.

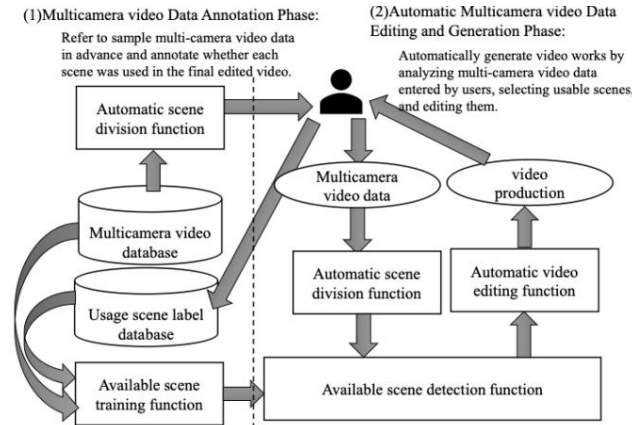


Fig. 1. Overview of our proposed method.

B. Automatic Scene Division Function

This function is the first step in both the training and generation. Its purpose is to segment continuous video footage into semantically consistent "scenes." This function processes each camera's footage independently, as we recognize that scene boundaries are often asynchronous across different camera angles. The process is as follows. First, we performed feature extraction. Instead of using low-level features, such as RGB histograms (which are sensitive to minor lighting changes), we use a pre-trained OpenAI CLIP model (Radford et al., 2021). CLIP embeds a video frame into a high-dimensional feature space, in which semantically similar frames are close together. This allows us to capture the content (e.g., "close-up of a soloist") rather than just the pixel values. This resulted in a rich embedding data vector for each frame. Then, clustering was performed. K-means clustering was applied to the sequences of the CLIP feature vectors. K-means groups temporally adjacent frames into clusters, where the frames within a cluster are semantically similar. This process effectively segments a continuous video stream into discrete semantically consistent scenes.

C. Available Scene Training Function

This function builds the predictive model, which is the "brain" of our editor. This was performed only during the Training Phase. First, for feature aggregation, for each scene identified in Section III.B, we aggregated all its frame-level CLIP embedding vectors by calculating their mean (average). This creates a single high-dimensional feature vector that represents an entire scene. Second, for labeling, this aggregated scene vector is then paired with the manually assigned label from the Usage Scene Label Database: 1 (adopted) or 0 (rejected). This dataset, consisting of (Scene Vector, Label) pairs, forms our training data. Third, for model training, this dataset was used to train a Support Vector Machine (SVM) classifier. As established in Section II.D, we chose SVM specifically because of its robustness on our extremely limited training set (33 scenes). We employed the Radial Basis Function (RBF) kernel, the default in scikit-learn, which is highly

effective for capturing the non-linear, complex, and implicit "rules" that constitute an editor's preferences.

D. Available Scene Detection Function

This function is used during the Generation Phase. The model trained as described in Section III was applied. C to a new, unseen footage. The input was a new scene (segmented by III.B) as its averaged CLIP feature vector. During inference, the trained SVM model analyzes this vector and determines whether it belongs to the "Adopted" or "Rejected" class. The output is a probability score (between 0.0 and 1.0) rather than a simple binary classification. We define this score as the "Adoptability Score," which indicates the model's confidence that the editor would adopt that scene. This score was a crucial input for the final editing function.

E. Automatic Video Editing Function

This function performs the final edit using the "Adoptability Scores" in a novel scene-to-frame selection logic. This logic is key to our framework and is designed to handle the challenge of differing (asynchronous) scene boundaries across multiple cameras. The algorithm is as follows: First, score propagation occurs: the Adoptability Score (from III.D) calculated for an entire scene (e.g., Scene A from Camera 1 has a score of 0.9) is assigned to every single frame belonging to that scene. This was performed for all the scenes on all the cameras. Second, a frame-by-frame competition was conducted. The system iterates through the master timeline and one frame (s) at a time. At each specific moment (e.g., $t=12s$), it looks at all available cameras (e.g., four cameras) and compares the scores of their respective frames at that exact moment. Third, final cut selection is performed: the frame from the camera with the highest Adoptability Score at that moment " t " is selected as the final cut for that moment. Fourth, a video was assembled. This selection process was repeated for the entire video duration. The "winning" frames are arranged in chronological order to assemble the final, coherent video work. This ensures that the final video is always composed of footage deemed most effective by the learned model.

IV. EXPERIMENT

We conducted experiments to validate the effectiveness of our proposed framework, focusing not on raw classification accuracy, but on the practical quality of the final autonomously generated video product.

A. Experimental Setting

The experiments were conducted on a workstation equipped with Google Colaboratory and running Python 3.12. The implementation environment consisted of the following major libraries: CLIP 1.0, opencv-python-headless 4.12.0.88, ipywidgets 7.7.1, matplotlib 3.10.0, numpy 2.0.2, pandas 2.2.2, requests 2.32.4, torch 2.8.0+cu126, torchvision 0.23.0+cu126, torchaudio 2.8.0+cu126, pillow 11.3.0, tqdm 4.67.1, joblib 1.5.2, scikit-learn 1.6.1, moviepy 1.0.3, scipy 1.16.2, statsmodels 0.14.5, and IPython 7.34.0. Standard Python libraries, such as os, io, random, and itertools, were also used. The dataset comprised multi-camera footage of a wind ensemble concert, a genre where visual subjects (e.g., soloists, conductors) and audio-visual synchronization are both important.

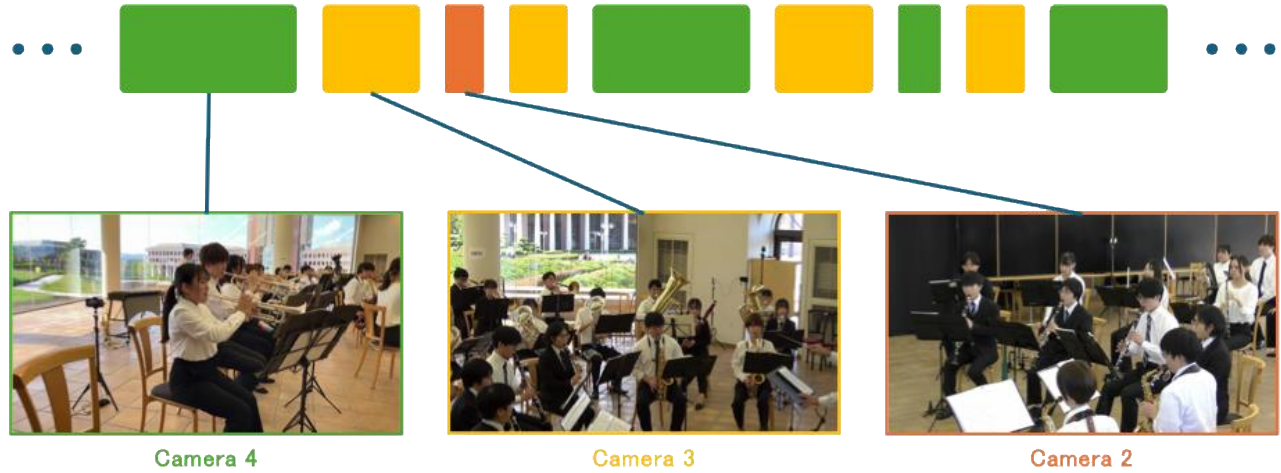


Fig.2. Example of editing results using this method from 2 seconds to 44 seconds.

Training Data: We used a 3 minute 28 second segment of footage from a single camera. This footage was segmented (using Method III.B) and hand-labelled by the researcher (editor) to create our ground truth. This process yielded an extremely limited training dataset of 33 scenes. The class distribution was nearly balanced, with 19 "Adopted" scenes and 14 "Rejected" scenes.

Test Data: We used a different 3 minute 28 second segment from the same concert. This segment was simultaneously filmed by four synchronized cameras, resulting in a total raw material of 13 min 52 s. These data were completely unseen by the trained model.

B. Evaluation Method : Usable Scene Ratio (USR)

The goal of this experiment was to evaluate the practical quality of the final autonomously generated video product. A standard "accuracy" metric (comparing to a human's single edit) is insufficient, as there can be multiple "correct" editing choices. Therefore, we define a new, product-oriented metric we call the "Usable Scene Ratio (USR)." This metric evaluates AI as an autonomous creator rather than a simple imitator. The evaluation procedure is as follows. First, the trained framework (Method III) autonomously generated a single, complete 3:28 video from the 4-camera test footage. Second, the human researcher (the same editor who provided the training labels) watched the final AI-generated video. Third, the researcher segments the AI-generated video into its constituent cuts (scenes) and performs a qualitative evaluation, labeling each cut as either "Usable" (i.e., the cut is acceptable and of sufficient quality to be in a final product) or "Not Usable" (i.e., the cut is a clear error, visually poor, or nonsensical). Finally, the USR is calculated as: (Number of "Usable" Scenes) / (Total Scenes in AI-generated video). In addition, to validate that the model's choices were not simply random, we performed a Wilcoxon signed-rank test. This statistical test compared the distribution of the model's assigned scores for the chosen scenes against a hypothetical random baseline.

C. Quantitative Results

The evaluation results, which form the primary claim of this study, are summarized in Table 1. The Usable Scene Ratio (USR) of the autonomously generated video was 79.66%. This key result indicates that approximately 80% of the cuts and scenes independently selected by the AI framework were deemed acceptable and of sufficient quality by a human expert editor. Furthermore, the Wilcoxon signed-rank test yielded a p-value of 1.37245×10^{-5} . This extremely low p-value ($p < 0.0001$) confirmed that the model selection was statistically significantly superior to random selection. These results demonstrate that, even with a training set of only 33 scenes, our framework is capable of autonomously producing a high-quality video that is far from random.

Table 1 Quantitative Evaluation Results

Evaluation Metric	Value
Usable Scene Ratio (USR)	79.66%
p-value (vs. Random)	1.37245×10^{-5}

D. Qualitative Analysis

Visual inspection of the editing results (see Fig. 2 for examples) provided deeper insights. Regarding success, the proposed method achieved transitions consistent with human intent in many scenarios. For example, in the segment from 02:31:00 to 02:37:00, the video composition was natural, relative to the subject's movements. This suggests the system successfully learned to identify visually important subjects (like a soloist or the conductor in the concert footage) from the CLIP features and correctly associated them with the "Adopted" label. Regarding failures and limitations, clear and critical limitations were observed. In some segments, the timing of the cut points was slightly misaligned with the rhythm of music. This is a crucial finding, because the dataset is a music concert. This failure can be attributed to the fact that our current framework relies exclusively on visual information (CLIP) and does not incorporate any audio features (e.g., beat, melody, and downbeat). Regarding novelty, interestingly, the system also generated some cut compositions that felt "valid" and "legitimate" even if the human editor would not typically have chosen them. This suggests that the model is not merely

memorizing, but generalizing, the editor's preference patterns, potentially discovering novel, creative editing choices.

V. DISCUSSION

This study proposes a limited-data-driven framework for autonomous multicamera editing. The results presented here are significant and warrant further discussion. The main finding of this study was that our method achieved a Usable Scene Ratio of 79.66%. We emphasize that this is not classification accuracy. This is the product-quality score. This result indicates that the final autonomously generated video was nearly 80% usable, despite the AI being trained on an extremely limited dataset of only 33 scenes. This, combined with the statistically significant p-value ($p < 0.0001$), strongly suggests that the success of the framework is not random.

The key question is why this framework functioned effectively despite the scarcity of data. We attribute this success to two primary factors and insist that the framework design is more important than the classifier itself. The first is the semantic feature representation (CLIP). The use of OpenAI CLIP (Method III.B) provided rich semantic representations of scenes. This allowed the SVM to learn meaningful patterns (e.g., "this is a good shot of a soloist") from the 33 examples, which would have been impossible with noisy, low-level RGB data. This semantic grounding allowed the model to make "intelligent" choices based on visual content, not just on pixels. The second is the scene-to-frame logic (Method III. E). The success is not just in the SVM, but also in the entire framework's architecture. Our novel scene-to-frame logic provides a robust and elegant way to translate abstract scene preferences (a single score) into concrete frame-by-frame editing decisions. This logic successfully navigates asynchronous multicamera footage to produce a coherent final product. This demonstrates that a well-designed framework can compensate for a lack of training data.

The limitations of this study are its success. Qualitative analysis (IV.D) revealed that the model's inability to consider audio information led to cuts misaligned with the musical rhythm of the concert. This confirms that for media such as concerts, sports, and interviews, a visual-only approach is insufficient. However, this limitation points directly to the most critical area for future work: Multimodal learning. The framework must be extended to integrate audio features (e.g., beat tracking, onset detection, Mel-Frequency Cepstral Coefficients (MFCCs), or even speech transcription) with visual CLIP features. This would allow the model to learn preferences like "cut on the downbeat" or "switch to the speaker." Furthermore, although 33 scenes were sufficient to demonstrate feasibility, scaling the training data was necessary to capture more complex and nuanced editing styles. By training on different editors, the model could potentially learn and reproduce "editing style modeling" —extracting and reproducing individual stylistic tendencies. This would elevate the system from a mere automated tool to an intelligent collaborative editing system.

VI. CONCLUSION

This study proposed a learning-based, limited-data-driven framework for the autonomous editing of multicamera footage.

Our core contribution is the integration of (1) semantic feature extraction (CLIP), (2) preference learning on extremely limited data (SVM), and (3) novel scene-to-frame selection logic that effectively generates a final video from asynchronous camera inputs.

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REFERENCES

- [1] S. Podlesnyy, "Towards data-driven automatic video editing," arXiv preprint arXiv:1907.07345, 2019.
- [2] D. M. Argaw, F. Caba Heilbron, J.-Y. Lee, M. Woodson, and I.-S. Kweon, "The Anatomy of Video Editing: A Dataset and Benchmark Suite for AI-Assisted Video Editing," arXiv preprint arXiv:2207.09812, 2022.
- [3] J.-H. Huang, C.-H. H. Yang, P.-Y. Chen, M.-H. Chen, and M. Worring, "Conditional modeling based automatic video summarization," arXiv preprint arXiv:2311.12159, 2023.
- [4] H.-D. Huynh-Lam, N.-P. Ho-Thi, M.-T. Tran, and T.-N. Le, "Enhancing video summarization with context awareness," arXiv preprint arXiv:2404.04564, 2024.
- [5] J. Tan, P. Yang, L. Chen, and H. Wang, "Temporal scene montage for self-supervised video scene boundary detection," ACM Trans. Multimedia Comput. Commun. Appl., 2024.
- [6] M. J. E. Brotons, F. J. Lucendo, J. Rodríguez-Juan, and J. García Rodríguez, "Shot boundary detection with 3D depthwise convolutions and visual attention," *Sensors*, vol. 23, 2023.
- [7] M. RajaSuguna, A. Kalaivani, and S. Anusuya, "The detection of video shot transitions based on primary segments using the adaptive threshold of colour-based histogram differences and candidate segments using the SURF feature descriptor," *Symmetry*, vol. 14, 2022.
- [8] E. Hato, "Temporal video segmentation using optical flow estimation," *Iraqi J. Sci.*, 2021.
- [9] Y. Hou, S. Gould, and L. Zheng, "Learning to select camera views: Efficient multiview understanding at few glances," arXiv preprint arXiv:2303.06145, 2023.
- [10] S. Majumder, T. Nagarajan, Z. Al-Halah, R. Pradhan, and K. Grauman, "Which viewpoint shows it best? Language for weakly supervising view selection in multi-view videos," arXiv preprint arXiv:2411.08753, 2024.
- [11] F. Lefevre, V. Bombardier, P. Charpentier, and N. Krommenacker, "Context-based camera selection from multiple video streams," *Multimedia Tools Appl.*, 2021.
- [12] A. Ranjeet, P. Kumar, K. C. Zagabattuni, and D. O. Rodgers, "Selecting a view of a multi-view video," US Patent US20170244959A1, 2016.
- [13] K. Host, M. Pobar, and M. Ivašić-Kos, "Analysis of movement and activities of handball players using deep neural networks," *J. Imaging*, vol. 9, 2023.
- [14] N. Liu, L. Liu, and Z. Sun, "Football game video analysis method with deep learning," *Computational Intelligence and Neuroscience*, 2022.
- [15] Z. Ruiye, "Volleyball training video classification description using the BiLSTM fusion attention mechanism," *Heliyon*, 2024.
- [16] P. Hu, N. Xiao, F. Li, Y. Chen, and R. Huang, "A reinforcement learning-based automatic video editing method using pre-trained vision-language model," arXiv preprint arXiv:2411.04942, 2024.
- [17] J. P. Schneider, P. S. Bisht, I. Chugunov, A. Kolb, M. Moeller, and F. Heide, "Neural atlas graphs for dynamic scene decomposition and editing," arXiv preprint arXiv:2509.16336, 2025.
- [18] S.-N. Chen, "Storyboard-based accurate automatic summary video editing system," *Multimedia Tools Appl.*, 2017.